

SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

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Experiment No.: 6 Name: Vansh Mahant PRN: 24070521038 Section: A Batch: A1

1. Experiment Title:

Regex & Profile Validator — a web page using HTML, CSS and JavaScript that extracts a student's profile details (name, roll number, email, phone, department) from raw text using regular expressions, validates the email and phone number formats, and performs text analysis (word/character count) and text transformations (lowercase conversion, string replacement).

2. Software/Tools Required:

Google Chrome (Web Browser), Visual Studio Code (Code Editor), Browser Developer Tools / Console, HTML, CSS, JavaScript

3. Experiment Program Code:

[index.html](#)

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Practical 9: Regex & Profile Validator</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <div class="container">
    <h2>Practical 9</h2>
    <p class="subtitle"></p>

    <!-- Input Section -->
    <div class="card">
      <h3>Input Text</h3>
      <label for="rawInput">Enter raw text containing student profile details:</label>
      <textarea id="rawInput"
        placeholder="Type or paste text here... (e.g., Student Name: Jane Doe, Roll
Number: 2024CS01, Email: jane@gmail.com, Phone: 9876543210, Department: Computer
Science)"></textarea>

      <div class="button-group">
        <button class="btn btn-secondary" onclick="loadSample()">Load Sample
Text</button>
        <button class="btn btn-primary" onclick="processText()">Process Text</button>
      </div>
    </div>

    <!-- Output Results Section -->
    <div id="resultsSection" class="results-container hidden">
      <!-- 1. Extracted Details & Validations -->
      <div class="card">
        <h3>1. Extracted Info & Validation</h3>
        <div class="detail-row">
          <span class="detail-label">Student Name:</span>
          <span class="detail-value" id="outName">-</span>
        </div>
        <div class="detail-row">

```

```

        <span class="detail-label">Roll Number:</span>
        <span class="detail-value" id="outRoll">-</span>
    </div>
    <div class="detail-row">
        <span class="detail-label">Email Address:</span>
        <span class="detail-value">
            <span id="outEmail">-</span>
            <span id="badgeEmail" class="badge">Pending</span>
        </span>
    </div>
    <div class="detail-row">
        <span class="detail-label">Phone Number:</span>
        <span class="detail-value">
            <span id="outPhone">-</span>
            <span id="badgePhone" class="badge">Pending</span>
        </span>
    </div>
    <div class="detail-row">
        <span class="detail-label">Department:</span>
        <span class="detail-value" id="outDept">-</span>
    </div>
</div>

<!-- 2. Text Analysis & Counting -->
<div class="card">
    <h3>2. Text Analysis</h3>
    <div class="detail-row">
        <span class="detail-label">Total Words:</span>
        <span class="detail-value" id="outWords">0</span>
    </div>
    <div class="detail-row">
        <span class="detail-label">Total Characters (excluding spaces):</span>
        <span class="detail-value" id="outChars">0</span>
    </div>
</div>

<!-- 3. Case Conversion & Replacement -->
<div class="card">
    <h3>3. Text Transformations</h3>
    <div class="transform-box">
        <label>Converted to Lowercase:</label>
        <div class="text-display" id="outLowercase">-</div>
    </div>
    <div class="transform-box">
        <label>Replaced Department ("Computer Science" → "Information
Technology"):</label>
        <div class="text-display" id="outReplaced">-</div>
    </div>
</div>
</div>

<script src="script.js"></script>
</body>
</html>

```

style.css

```

/* Basic layout and typography */
body {
    font-family: Arial, sans-serif;
    margin: 30px auto;
    max-width: 600px;
    line-height: 1.6;
    background-color: #fafafa;
    color: #333;
}

.container {

```

```
padding: 0 10px;
}

h2 {
  color: #2c3e50;
  margin-bottom: 5px;
  text-align: center;
}

.subtitle {
  text-align: center;
  color: #666;
  margin-top: 0;
  margin-bottom: 25px;
}

/* Card/Section box */
.card {
  border: 1px solid #ccc;
  padding: 15px;
  margin-bottom: 20px;
  border-radius: 5px;
  background-color: #ffffff;
}

h3 {
  margin-top: 0;
  color: #2980b9;
  border-bottom: 1px solid #eee;
  padding-bottom: 5px;
  margin-bottom: 12px;
}

label {
  display: block;
  margin-top: 10px;
  font-weight: bold;
}

textarea {
  width: 100%;
  height: 100px;
  padding: 8px;
  margin: 8px 0;
  box-sizing: border-box;
  border: 1px solid #ccc;
  border-radius: 4px;
  resize: vertical;
}

/* Button styles */
.button-group {
  margin-top: 10px;
}

.btn {
  padding: 8px 15px;
  font-weight: bold;
  border: none;
  border-radius: 3px;
  cursor: pointer;
  margin-right: 5px;
}

.btn-primary {
  background-color: #2ecc71;
  color: white;
}
```

```
.btn-primary:hover {
    background-color: #27ae60;
}

.btn-secondary {
    background-color: #3498db;
    color: white;
}

.btn-secondary:hover {
    background-color: #2980b9;
}

/* Extracted details rows */
.detail-row {
    padding: 8px 0;
    border-bottom: 1px solid #eee;
}

.detail-row:last-child {
    border-bottom: none;
}

.detail-label {
    font-weight: bold;
    color: #555;
    display: inline-block;
    width: 150px;
}

.detail-value {
    color: #000;
}

/* Badges for validation */
.badge {
    font-weight: bold;
    padding: 2px 6px;
    border-radius: 3px;
    font-size: 11px;
    margin-left: 8px;
    display: inline-block;
}

.badge-valid {
    background-color: #2ecc71;
    color: white;
}

.badge-invalid {
    background-color: #e74c3c;
    color: white;
}

.badge-pending {
    background-color: #95a5a6;
    color: white;
}

/* Transformations and outputs */
.transform-box {
    margin-top: 10px;
}

.text-display {
    font-family: monospace;
    font-size: 13px;
    margin-top: 5px;
    padding: 10px;
}
```

```

background-color: #f5f5f5;
border-left: 4px solid #3498db;
word-break: break-all;
white-space: pre-wrap;
}

/* Visibility classes */
.hidden {
  display: none;
}

```

script.js

```

// Load a sample student profile text into the textarea
function loadSample() {
  const sampleText = `Student Name: Vansh Mahant
Roll Number: 24070521038
Email Address: vanshmahant3@gmail.com
Phone Number: 8446936170
Department: Computer Science`;

  document.getElementById('rawInput').value = sampleText;
}

// Main logic to process text, run regex, and validate details
function processText() {
  const rawText = document.getElementById('rawInput').value.trim();

  if (!rawText) {
    alert("Please enter some text or load the sample text first!");
    return;
  }

  // 1. Extract Details using Regular Expressions
  const nameMatch = rawText.match(/(?:Student\s+)?Name:\s*([^\n\r,]+)/i);
  const rollMatch = rawText.match(/(?:Roll\s*(?:Number|No)?)?:\s*([^\n\r,]+)/i);
  const emailMatch = rawText.match(/(?:Email(?:\s*Address)?)?:\s*([^\n\r\s,]+)/i) ||
rawText.match(/[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}/);
  const phoneMatch = rawText.match(/(?:Phone(?:\s*Number)?|Contact):\s*([^\n\r\s,]+)/i) ||
rawText.match(/\b\d{10}\b/) || rawText.match(/\b\d+\b/);
  const deptMatch = rawText.match(/(?:Department|Dept):\s*([^\n\r,]+)/i);

  const name = nameMatch ? nameMatch[1].trim() : "Not Found";
  const roll = rollMatch ? rollMatch[1].trim() : "Not Found";
  const email = emailMatch ? (emailMatch[1] ? emailMatch[1].trim() : emailMatch[0].trim()) :
"Not Found";
  const phone = phoneMatch ? (phoneMatch[1] ? phoneMatch[1].trim() : phoneMatch[0].trim()) :
"Not Found";
  const department = deptMatch ? deptMatch[1].trim() : "Not Found";

  // 2. Validate Email Address using Regular Expression
  const emailRegex = /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/;
  const isEmailValid = email !== "Not Found" && emailRegex.test(email);

  // 3. Validate Phone Number contains exactly 10 digits
  const phoneRegex = /^\d{10}$/;
  const isPhoneValid = phone !== "Not Found" && phoneRegex.test(phone);

  // 4. Counts
  // Count Words (split by whitespace)
  const wordsArray = rawText.split(/\s+/).filter(word => word.length > 0);
  const wordCount = wordsArray.length;

  // Count Characters excluding spaces
  const charCount = rawText.replace(/\s/g, '').length;

  // 5. Convert text to lowercase
  const lowercaseText = rawText.toLowerCase();

```

```

// 6. Replace "Computer Science" with "Information Technology"
const replacedText = rawText.replace(/Computer\s+Science/gi, "Information Technology");

// 6. Display all extracted information in the UI
document.getElementById('outName').textContent = name;
document.getElementById('outRoll').textContent = roll;
document.getElementById('outEmail').textContent = email;
document.getElementById('outPhone').textContent = phone;
document.getElementById('outDept').textContent = department;

// Update Email Validation Badge
const emailBadge = document.getElementById('badgeEmail');
if (email === "Not Found") {
    emailBadge.textContent = "Missing";
    emailBadge.className = "badge badge-invalid";
} else if (isEmailValid) {
    emailBadge.textContent = "Valid";
    emailBadge.className = "badge badge-valid";
} else {
    emailBadge.textContent = "Invalid";
    emailBadge.className = "badge badge-invalid";
}

// Update Phone Validation Badge
const phoneBadge = document.getElementById('badgePhone');
if (phone === "Not Found") {
    phoneBadge.textContent = "Missing";
    phoneBadge.className = "badge badge-invalid";
} else if (isPhoneValid) {
    phoneBadge.textContent = "Valid";
    phoneBadge.className = "badge badge-valid";
} else {
    phoneBadge.textContent = "Invalid";
    phoneBadge.className = "badge badge-invalid";
}

// Update Counts
document.getElementById('outWords').textContent = wordCount;
document.getElementById('outChars').textContent = charCount;

// Update Transformations
document.getElementById('outLowercase').textContent = lowercaseText;
document.getElementById('outReplaced').textContent = replacedText;

// Show the results section
const resultsSec = document.getElementById('resultsSection');
resultsSec.classList.remove('hidden');
}

```

4. Output:

PRN: 24070521038 | Name: Vansh Mahant | File Path: /practical6/index.html — Initial state

Practical 9

Input Text

Enter raw text containing student profile details:

Type or paste text here... (e.g., Student Name: Jane Doe, Roll Number: 2024CS01, Email: jane@gmail.com, Phone: 9876543210, Department: Computer Science)

Load Sample Text

Process Text

PRN: 24070521038 | Name: Vansh Mahant | File Path: /practical6/index.html — After loading sample text and processing

Practical 9

Input Text

Enter raw text containing student profile details:

```
Student Name: Vansh Mahant  
Roll Number: 24070521038  
Email Address: vanshmahant3@gmail.com  
Phone Number: 8446936170  
Department: Computer Science
```

Load Sample Text

Process Text

1. Extracted Info & Validation

Student Name: Vansh Mahant

Roll Number: 24070521038

Email Address: vanshmahant3@gmail.com Valid

Phone Number: 8446936170 Valid

Department: Computer Science

2. Text Analysis

Total Words: 16

Total Characters
(excluding
spaces): 128

3. Text Transformations

Converted to Lowercase:

```
student name: vansh mahant  
roll number: 24070521038  
email address: vanshmahant3@gmail.com  
phone number: 8446936170  
department: computer science
```

Replaced Department ("Computer Science" → "Information Technology"):

```
Student Name: Vansh Mahant  
Roll Number: 24070521038  
Email Address: vanshmahant3@gmail.com  
Phone Number: 8446936170  
Department: Information Technology
```

5. Result/Conclusion:

A Regex & Profile Validator web page was successfully designed and implemented using HTML, CSS and JavaScript. The program uses regular expressions to extract the student's name, roll number, email, phone number and department from raw text, validates the email format and confirms the phone number is exactly 10 digits, and

performs word/character counting along with lowercase conversion and department string replacement. This experiment reinforced the concepts of regular expressions, string parsing, pattern matching, and text transformation methods in JavaScript.

Upload the Soft copy as well as complete code on GitHub link is mandatory.

File name: Practical 6_24070521038_Vansh Mahant_A_A1